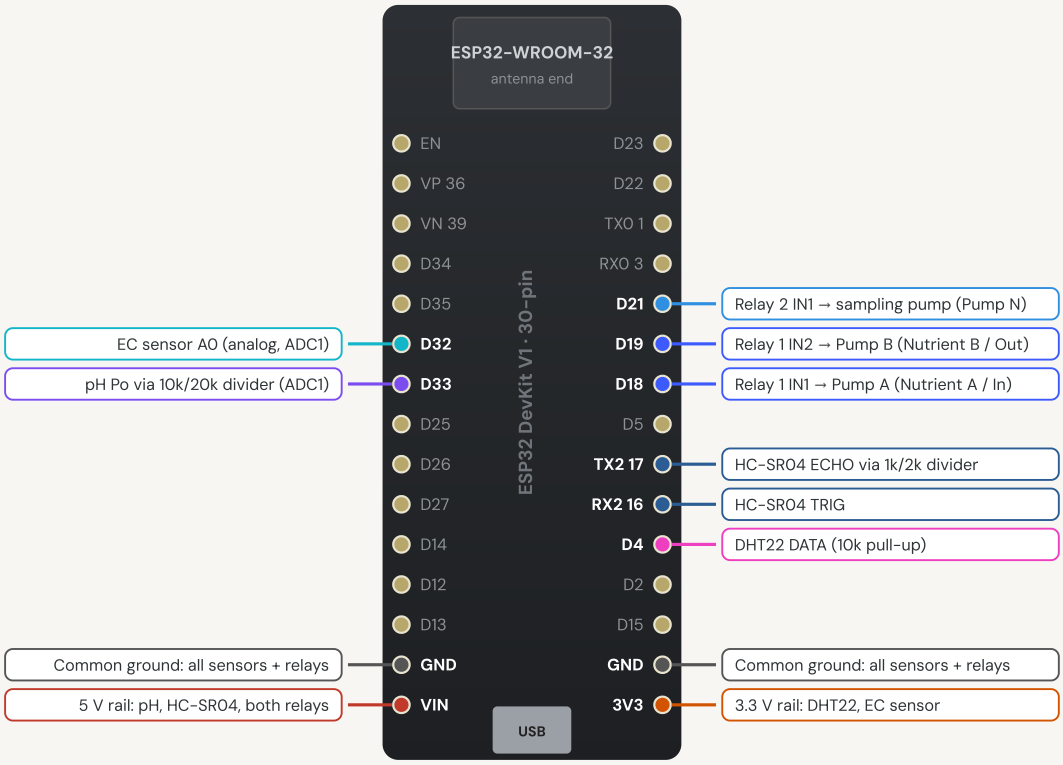
02

# ESP32 Pin Configuration

Which ESP32 pin goes to which sensor and relay, how each one is powered, and the two voltage dividers that keep 5 V away from the 3.3 V pins. Source of truth: `firmware/Full_code/Full_code.ino`.




Pin positions follow the common DOIT ESP32 DevKit V1 (30-pin) board. On a 38-pin board the GPIO numbers are the same but the positions differ, so wire by the GPIO number printed on the board.

## Pin assignment

| Device | Pin | ESP32 pin | Power | Notes |
|---|---|---|---|---|
| **DHT22** | `DATA` | `GPIO4 (D4)` | 3.3 V | 10k pull-up if the module has none |
| **EC sensor (EC V2.1)** | `A0` | `GPIO32 (D32)` | 3.3 V | ADC1 · T1 not connected |
| **pH sensor (PH-4502C)** | `Po` | `GPIO33 (D33)` | 5 V | ADC1 · **10k / 20k divider required** |
| **HC-SR04** | `TRIG` | `GPIO16 (D16)` | 5 V | 3.3 V trigger is enough |
| **HC-SR04** | `ECHO` | `GPIO17 (D17)` | 5 V | **1k / 2k divider required** (5 V echo pulse) |
| **Relay 1 · Pump A (In)** | `IN1` | `GPIO18 (D18)` | 5 V | Active LOW · GND shared with ESP32 |
| **Relay 1 · Pump B (Out)** | `IN2` | `GPIO19 (D19)` | 5 V | Active LOW |
| **Relay 2 · Sampling pump** | `IN1` | `GPIO21 (D21)` | 5 V | Active LOW · relay 2 IN2 spare |

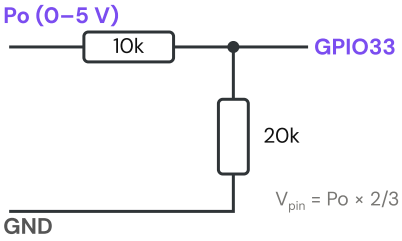
> **Never put more than 3.3 V on an ESP32 pin.** Keep the pH and EC probes in separate containers when testing; they interfere with each other.

PROTECTING THE 3.3 V PINS

# Voltage dividers

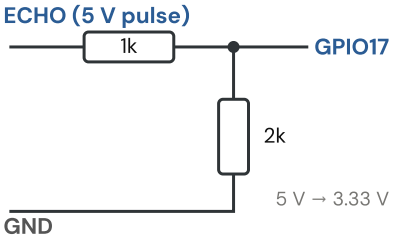
The ESP32 pins take at most 3.3 V. Two modules run on 5 V and can output more than that, so each gets a resistor divider.

### pH output (Po → GPIO33)




Firmware multiplies back by `PH_DIVIDER 1.5`. A Po of 4.6 V is about 3.07 V at the pin (`PH_ADC_LIMIT`).

### HC-SR04 echo (ECHO → GPIO17)




TRIG is driven straight from GPIO16. Each level reading is the median of 5 pings, corrected for air temperature.

## Power rails

| Rail | Powers | Why |
|---|---|---|
| **3.3 V** | DHT22, EC sensor | Output already within the ESP32 range |
| **5 V** | pH module, HC-SR04, relay module 1 and 2 | Modules need 5 V to work properly |
| **GND** | Everything, shared with the ESP32 | Relays and sensors need a common reference |

## Why these pins

- **Analog inputs on ADC1** (GPIO32, GPIO33). ADC2 pins cannot be read while Wi-Fi is active, so ADC1 keeps the design ready for that.
- **No strapping pins used** (GPIO0, 2, 5, 12, 15). They decide the boot mode, and a sensor on them can stop the ESP32 from starting.
- **Relays default OFF.** The firmware sets the relay pins to OFF *before* making them outputs, so the pumps do not kick on at boot (`RELAY_ACTIVE_LOW true`: LOW = ON).

## Firmware settings (Full_code.ino)

| Setting | Value | Meaning |
|---|---|---|
| `EC_MIN / EC_MAX` | 1.2 / 1.8 mS/cm | EC target band for pump auto control |
| `PH_MIN / PH_MAX` | 5.5 / 6.5 | pH target band for pump auto control |
| `TEMP_MIN / TEMP_MAX` | 25 / 36 °C | Air-temperature status band (report only) |
| `LOW_LEVEL_PCT · LOW_CONFIRM` | 20 % · 3 | Tank low after 3 low readings in a row → all pumps off |
| `SAMPLE / PUMP_START_DELAY` | 30 s / 60 s | Wait after boot before the sampling pump / Pump A+B start |
| `SEND_INTERVAL · DHT_INTERVAL` | 2 s · 4 s | JSON line every 2 s over USB serial at 115200 baud |